NAME: ____________________ ID: __________

CSC 343 H5F — VERSION A

Day Class 01

Instructor: Michael Liut

DURATION: 2 hours

University of Toronto Midterm Examination October 31st, 2018

Read all instructions before starting the exam

This test paper includes 15 pages and a total of 18 questions; 13 true-or-false questions, 1 fill-in-the-blank question, and 4 questions requiring a written answer. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

question(s)	mark	out of
1 – 14		20
15		10
16		10
17		10
18		3
misc		n/a
total		53

Special Instructions :

1. The answers to the written questions (Questions 15-18) must be answered in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked.
2. Read all questions before starting the exam – some are easier than others.
3. The True-or-False section requires you to illustrate the validity by an example, or disprove the validity by a counter-example. If your justification is proof by definition, this is perfectly valid!
4. **This is a closed book exam.** No memory aids, notes, calculators, or, textbooks of any kind are allowed during the test. The use of any electronic device is strictly prohibited. Failure to comply will result in your immediate dismissal from the examination and a grade of 0.
5. Students are not allowed to be involved in any communication of any kind. Questions concerning this exam must be brought to the attention of the invigilator(s) or the instructor. Any attempt of alternative communication will be considered a case of academic dishonesty.
6. No unauthorized scrap paper is allowed to be used. The invigilator(s) will supply every student with needed scrap paper when asked. An additional mark will be awarded to students who circle this line; do not look around, just do it.
7. Documents to be returned: this questionnaire and all scrap paper if used. All of these documents must bear the student's name and number. Only the face page of the questionnaire needs to bear the student's name and number.

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Question 1 – 13 are True-or-False questions. You are required to illustrate the validity by an example, or disprove the validity by a counter-example.

For each question only select one answer (i.e. True or False). To select an answer, you must circle the word "True" or the word "False". The negative marking scheme is not used. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. For full marks you must justify (prove/disprove) your answer.

Question 1 [1.5 marks]
A weak-relation $W(ForeignKey, Attribute_2, ..., Attribute_n)$, with `n`-attributes, has a $ForeignKey$ which references the $PrimaryKey$ from strong-relation $S(\underline{PrimaryKey}, Attribute_2, ..., Attribute_n)$, with `n`-attributes. This means, minimally, that one of W's attributes must always also be part of the key for W (i.e. W's key would be: $\{ForeignKey, Attribute_i\}$, where $Attribute_i$ is an arbitrary attribute belonging to W).

A. True.
B. False.

Question 2 [1.5 marks]
A relation is made up of two components: (1) an instance (i.e. a table defined by its cardinality and degree/arity); and (2) a schema (i.e. the relation name and the type of each column).

A. True.
B. False.

Question 3 [1.5 marks]
Physical data independence separates the physical (refers to the hardware-level/system design) and conceptual schemas. Logical data independence separates the conceptual schema from the external schema, which depends on relations in the conceptual schema.

A. True.
B. False.

Question 4 [1.5 marks]
Given two relations R and S, the Cartesian Product is denoted $R \times S$, where R and S are sets: $R \times S = \{(r, s) : (r \in R) \text{ and } (s \in S)\}$. Thus, in general, $(R \times S) = (S \times R)$ (i.e. the Cartesian Product operation is commutative).

A. True.
B. False.

Question 5 [1.5 marks]
A domain constraint expresses conditions on the values of each tuple, independent of other tuples. A tuple constraint is a domain constraint that involves a single attribute.

A. True.
B. False.

Question 6 [1.5 marks]
Triggers are more commonly used to maintain database consistency, alert us to unusual events, allow us to impose flexible constraints, and can even generate a log of events to support auditing and security checks. Where as *Constraints* prevent inconsistency by any kind of statement, afford the DBMS the opportunity to optimize, and are easier to understand than Triggers.

A. True.
B. False.

Question 7 [1.5 marks]
With respect to *Bag Law*, it can be said that the Idempotent Law[1], in general, for union does hold for bags (i.e. given some relation R, then $R \cup R = R$).

A. True.
B. False.

[1] real definition: an element x of some operator $\cdot$ is said to be idempotent if $x \cdot x = x$. If all elements are idempotent with respect to $\cdot$, then $\cdot$ is called idempotent. i.e. $\forall x$, $x \cdot x = x$ then idempotent law holds for $\cdot$.

Question 8 [**1.5 marks**]
Suppose you have two relations `R` and `S`, where a tuple of `R` that has no tuple of `S` with which it joins is said to be dangling. Assuming that we take `R` and JOIN it to `S` (i.e. $R \bowtie S$), then to preserve[2] each tuple in `R` we can perform any of the following JOINs (independently): "OUTER", "RIGHT", "INNER", or "CROSS"[3].

A. True.
B. False.

Question 9 [**1.5 marks**]
Given two of relations `R` and `S`, in the context of data modeling, there are only three degrees of cardinality that can be between them: (1) one-to-one $(1 : 1)$; (2) one-to-many $(1 : M)$; and (3) many-to-many $(M : N)$.

A. True.
B. False.

Question 10 [**1.5 marks**]
Let's assume that we are working with subset of a banking schema. Given two of the entities: `Customer` and `Loan`, and their relationship `Borrower`, it is <u>fair to assume</u> that there is a partial participation between `Customer` —— `Borrower` and <u>clear to assume</u> a total participation between `Borrower` ═══ `Loan`. [4]

A. True.
B. False.

[2] your join may pad values with NULL, but cannot lose any information from `R`'s tuples
[3] <u>real definition</u>: CROSS JOIN is equivalent to taking the Cross Product
[4] <u>real definition</u>: —— indicates partial participation and ═══ indicates total participation

Question 11 [**1.5 marks**]
The Database Administrator (DBA) is responsible for ensuring these critical tasks: (1) the security and authorization of the data on the DBMS; (2) the data's availability and recovery from any failure; and (3) database tuning, which will evolve over time. The DBA, however, is not responsible for the design of the conceptual and physical schemas.

A. True.
B. False.

Question 12 [**1.5 marks**]
Cardinality, with respect to uniqueness of data, says that something has low cardinality if it has a low number of duplicates in the data and high cardinality if the data has a high number of duplicates.

A. True.
B. False.

Question 13 [**1.5 marks**]
In an entity-relationship diagram, the ISA hierarchy is similar to the parent-child relationship in object oriented programming and is organized in a tree-like structure. Therefore, it is <u>fair to assume</u> that if a parent relation `P` has three children relations `X`, `Y`, and `Z`, then any member of `P` will not be members of more than one child (e.g. $(\exists a \mid (a \in P)\ and\ (a \in X) \implies (a \notin Y)\ and\ (a \notin Z)))$.

A. True.
B. False.

Question 14 [**0.5 marks**]

Fill in the blank: the name[5] of one of my Teaching Assistants is ______________________.

[5] first name, last name, and/or common name will be taken as correct

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Questions 15 – 18 are questions that require a written answer. The answers are to be written in the space provided below each question.

Question 15 [10 marks]
Consider the relations: `Students`, `Faculty`, `Courses`, `Rooms`, `Enrolled`, `Teaches`, and `MeetsIn`; more precisely defined as:

`Students`(sid: string, name: string, login: string, age: integer, gpa: real)
`Faculty`(fid: string, fname: string, sal: real)
`Courses`(cid: string, cname: string, credits: integer)
`Rooms`(rno: integer, address: string, capacity: integer)
`Enrolled`(sid: string, cid: string, grade: string)
`Teaches`(fid: string, cid: string)
`MeetsIn`(cid: string, rno: integer, time: string)

(a) [5 marks] Create an entity-relationship diagram that denotes the relations given above.

(b) [3 marks] List all the foreign key constraints among these relations.

(c) [2 marks] Give an example of two possible domain constraints on the relations above.

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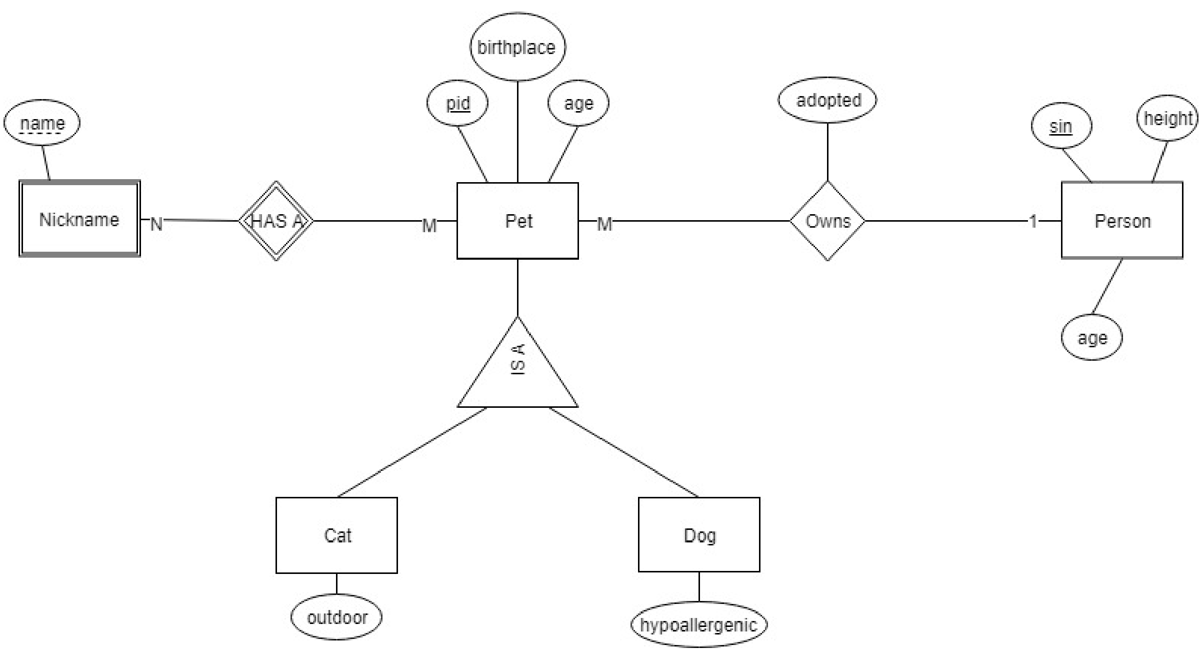
Question 16 [10 marks]
Given the following Entity-Relationship Diagram (ERD), in Chen's Notation below, you must translate the ERD to its appropriate Data Definition Language (DDL).

Requirements: (1) any attribute requiring a "Yes" or "No" answer should be reflected as a boolean value, all others must be reasonably determined (type and size); (2) you must enforce that *a Dog cannot be a Cat* and *a Cat cannot be a Dog*; (3) no intra-relational constraints need be created for any relation; (4) all inter-relational constraints must be enforced; and (5) each time an entry is removed/deleted all of its associated entries are also removed/deleted.



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Question 17 [10 marks]
Given the following DDL:

```
CREATE TABLE Table_1 (
      A int
      B int
      C int
      PRIMARY KEY(A)
);
```

```
CREATE TABLE Table_2 (
      D int
      E int
      F int
      PRIMARY KEY(D,F)
);
```

```
delimiter $$
CREATE TRIGGER trig AFTER INSERT ON Table_1
      FOR EACH ROW
          BEGIN
                INSERT INTO Table_2(F, E, D) VALUES(0, NEW.A, NEW.A + 5)
          END;
$$ delimiter;
```

For each question below, assume that the DML given is run on a clean version of the DDL (above); that is, each part does not affect the other. With all the information provided, your task is to provide the output[6] of the following DML executions:

(a) [2 marks] `INSERT INTO Table_2 VALUES(1, 7, 9);`

(b) [3 marks] `INSERT INTO Table_1 VALUES(1, 2, 3);`

(c) [3 marks]
```
INSERT INTO Table_1(B,A) VALUES(4, 5);
INSERT INTO Table_1(A,C) VALUES(4, 5);
```

(d) [2 marks]
```
INSERT INTO Table_1(B,A) VALUES(4, 5);
INSERT INTO Table_1(B,A) VALUES(10, 9);
INSERT INTO Table_2(F,E,D) VALUES(0, 0, 24-10);
INSERT INTO Table_1(A,B) VALUES(10, 9);
```

[6] you may opt to draw the tables content or write out what each table contains. In the event of an error, you must indicate what that error is and what caused it.

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Question 18 [**3 marks**]
Consider the schema below about movies and actors (keys are underlined):

Actor(<u>aid</u>, name)

Movie(<u>mid</u>, title, year, genre, length)

Roles(<u>aid</u>, <u>mid</u>, character)

You are given an SQL query which finds all actors who did not star in any movie (i.e. play a character) in the year 2018 and only returns the actor's name and actor ID:

```
SELECT aid, name
FROM   Actor
WHERE  aid   NOT IN
             (SELECT roles.aid
              FROM   Actor, Roles, Movie
              WHERE  year = 2018 AND actor.aid = roles.aid
                     AND roles.mid = movie.mid);
```

Your task is to translate the SQL query (above) into Relational Algebra.

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